Project Proposal

Enhancing Clinical Text Annotation with Fine-Tuned NLP Models

Group Information:

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Introduction

Healthcare systems generate vast amounts of unstructured clinical text daily, such as patient notes, discharge summaries, and test results. Extracting structured insights from this data is essential for improving decision-making, reducing administrative tasks, and enhancing patient care. Natural Language Processing (NLP) models, such as MedCAT, offer medical entity recognition and linking tools, but their performance often requires domain-specific adaptation to ensure accuracy and relevance.

Existing NLP models may lack precision when applied to specific clinical scenarios, particularly when recognizing diseases or capturing contextual information like negation or certainty. This project focuses on adapting and fine-tuning state-of-the-art models to bridge this gap and improve their applicability in healthcare settings.

Improved clinical text annotation can lead to more effective decision support, better patient outcomes, and operational efficiency in healthcare systems.

Problem Statement

Current medical NLP tools face challenges when applied to clinical text with diverse terminology and context-specific requirements. Recognizing entities and understanding nuanced attributes such as certainty or negation remain key areas for improvement.

Objectives and Goals

Objectives:

1. Fine-tune an NLP model for clinical text annotation.
2. Enhance the model's ability to recognize diseases and related concepts.
3. Integrate contextual understanding to improve annotation accuracy.

Expected Outcomes:

- A fine-tuned NLP model capable of high-accuracy clinical entity annotation.
- Contextual information, such as certainty or negation, is incorporated into annotations.
- Documentation and example outputs showcasing enhanced annotation performance.

Proposed Solution

Approach:

The project leverages an open-source NLP framework and enhances its performance through fine-tuning and configuration. Key steps include:

1. **Model Training**: Train and fine-tune the model on clinical text for improved entity recognition and linking.
2. **Contextual Understanding**: Integrate additional components for capturing meta-information such as certainty or negation.
3. **Optimization**: Adjust parameters (e.g., learning rate, dropout) to ensure optimal performance.

4. **Evaluation**: Assess performance through validation metrics and example annotations.

Technologies, Tools, and Frameworks:

- **MedCAT**: For entity recognition and linking.
- **MetaCAT**: For adding contextual understanding.
- **Python**: For scripting and data processing.

Scope of Work

In Scope:

- Fine-tuning the NLP model to improve clinical text annotation.
- Adding and training meta-annotation capabilities for richer contextual understanding.
- Outputting results demonstrating improved annotation quality.

Out of Scope:

- Large-scale deployment of the model.
- Integration with live clinical systems.

Expected Challenges

1. **Model Adaptation**: Ensuring the model adapts well to clinical language variations.
2. **Context Understanding**: Capturing subtle contextual cues like certainty or negation accurately.
3. **Computational Constraints**: Balancing model complexity with available resources.